

IT UNIVERSITY OF COPENHAGEN

Predicting Insurance Claim Frequency Using Machine Learning

Elena Linares Leal
el@itu.dk

Ruben Coloma Garcia
ruco@itu.dk

Course: Machine Learning (Autumn 2025)
Course Code: BSMAL11KU
Course Manager: Jonas Juul

GitHub repository:
Machine Learning Final Project

15-12-2025

1 Introduction

Insurance claim frequency modeling plays a central role in the insurance industry, as it directly affects premium calculation, risk assessment, and portfolio management. Insurers rely on accurate estimates of the expected number of claims in order to price policies fairly and ensure financial stability. Even small inaccuracies in claim frequency estimation can lead to substantial financial consequences when applied across large portfolios.

Traditionally, insurance claim frequency has been modeled using relatively simple statistical methods that assume linear relationships between policy characteristics and claim occurrence. These models are easy to interpret and computationally efficient, but they often struggle to capture complex patterns present in real-world insurance data, typically noisy, highly imbalanced, and dominated by policies with no claims. In recent years, advances in machine learning have motivated the use of more flexible models that can learn non-linear relationships directly from data. While such models have shown promising results on tabular datasets, their increased complexity raises questions about whether they offer meaningful advantages over simpler approaches in practical insurance applications.

In this project, we investigate whether machine learning models can effectively predict insurance claim frequency using a real-world dataset. The target variable represents the number of claims associated with an insurance policy, while the input features describe policy and vehicle characteristics, as well as an exposure variable representing time at risk.

Our goal is to compare different modeling approaches with varying levels of complexity. We implement a decision tree and a feed-forward neural network both from scratch and using established machine learning libraries, and we additionally apply a gradient boosting model using LightGBM with a Poisson loss function tailored for count data. By comparing these models, we aim to assess their predictive performance, robustness, and suitability for insurance claim frequency modeling in practice.

2 Data Analysis

The dataset used in this project contains information about automobile insurance policies and their associated claims history over a one-year period. Each observation corresponds to a single insurance policy and includes policy, vehicle, and driver characteristics, along with the

number of claims observed during the exposure period. The dataset consists of approximately 680,000 entries and is split into a predefined training set (80%) and test set (20%), which are used consistently throughout the project.

2.1 Dataset Description

The target variable in this study is the number of claims (*ClaimNb*) associated with a policy. This variable represents count data and is highly skewed, with the majority of policies having zero claims. Such imbalance is common in insurance data and makes prediction challenging, as models must distinguish rare claim events from a large number of non-claim observations.

Approximately 5% of policies in both the training and test sets are associated with at least one claim:

- Train *ClaimNb*: 5.31222%
- Test *ClaimNb*: 5.37451%

Several explanatory variables describe characteristics of the vehicle and the driver, including vehicle age, vehicle power, driver age, fuel type, and region of residence. The dataset also includes an *Exposure* variable, which represents the fraction of a year during which a policy was observed.

Figure 1 illustrates the empirical distributions of the main numerical variables in the training dataset.

Figure 1 highlights several key characteristics of the data. The target variable *ClaimNb* is extremely right-skewed, with the vast majority of policies having zero claims and only a small fraction exhibiting multiple claims. This confirms the strong zero-inflation present in the dataset.

Several explanatory variables also exhibit skewed distributions. Vehicle age and bonus-malus scores are concentrated at lower values with long right tails, while population density shows extreme skewness, reflecting large differences between rural and urban areas. In contrast, driver age follows a more symmetric distribution. These heterogeneous distributions suggest that linear modeling assumptions may be restrictive and motivate the use of flexible, non-linear models.

2.2 Preprocessing and Filtering

An initial inspection of the data revealed that while most exposure values are close to one year, a subset of policies has exposure values exceeding the expected observation period. In this project, exposure values are capped

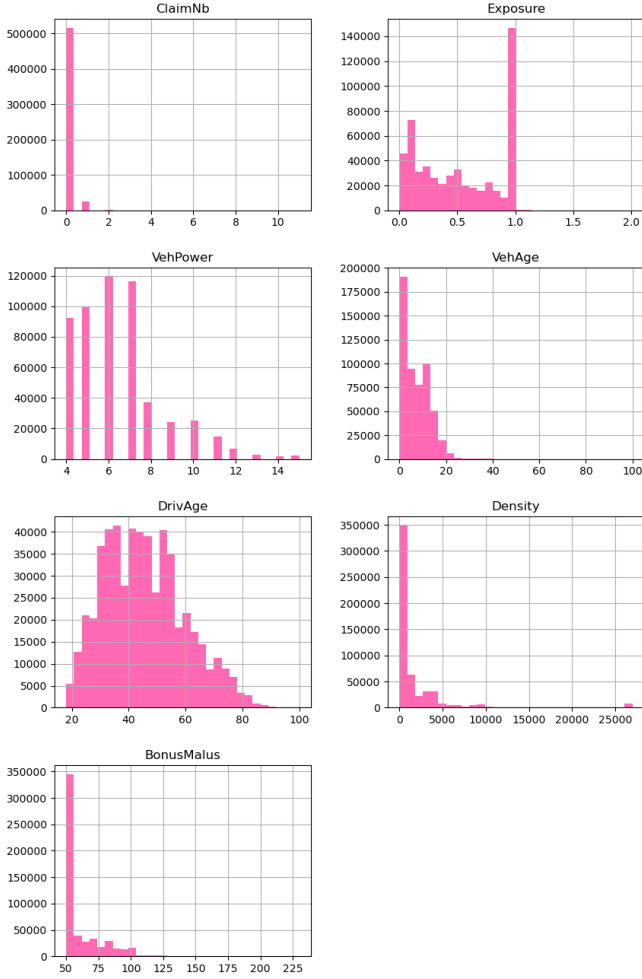


Figure 1: Visualization of the numerical variables.

at one year, and an additional indicator feature is introduced to flag policies with unusually large exposure. This approach preserves information about atypical exposure while preventing extreme values from disproportionately influencing the models.

Categorical variables are encoded numerically, and missing values are handled using simple and robust strategies. Overall, the dataset presents a realistic insurance modeling challenge due to its heterogeneity, noise, and strong class imbalance.

2.3 Exploratory Analysis and Observations

To gain an overview of the dataset, dimensionality reduction using Principal Component Analysis (PCA) and clustering methods were applied. These visualizations suggest that insurance policies do not separate into clearly distinct groups based on the available features alone.

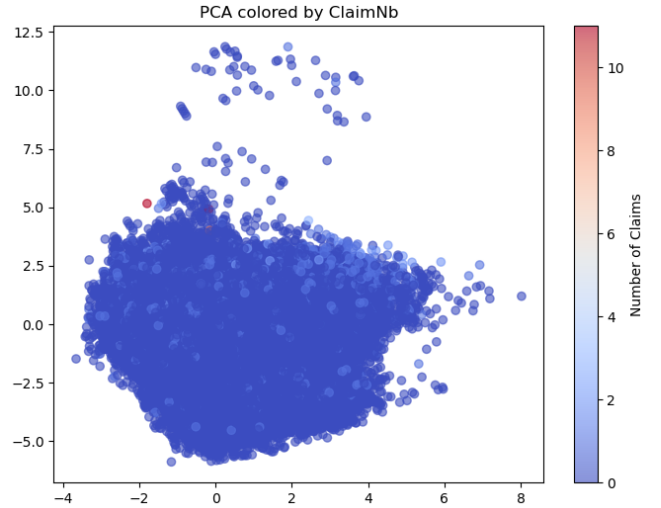


Figure 2: PCA projection of the training data, colored by the number of claims (*ClaimNb*). Most observations cluster around zero claims, with no clear separation between higher-claim policies.

The PCA projection reveals a strong overlap between policies with different claim counts, indicating that claim risk cannot be separated using simple linear projections alone, as shown in Figure 2.

Applying K-means clustering in the PCA-reduced space further supports this observation. The resulting clusters exhibit substantial overlap and do not correspond to distinct levels of claim risk, suggesting that unsupervised clustering alone is insufficient to separate high-risk and low-risk policies (Figure 3).

3 Methods

3.1 Target Variable and Evaluation Metrics

In this project, claims risk is defined as the expected number of insurance claims associated with a policy during the observed exposure period. The target variable is therefore the number of claims, denoted as *ClaimNb*.

Although claim frequency is often expressed as the ratio between the number of claims and exposure ($ClaimNb/Exposure$), this project models the expected number of claims directly. Exposure is handled explicitly as an input feature through capping, transformation, and indicator variables, allowing the models to learn the effect of time at risk without requiring an explicit normalization of the target. Since exposure is a maximum of one year, the predicted claim counts can be interpreted as annualized expected claims, making *ClaimNb* a rea-

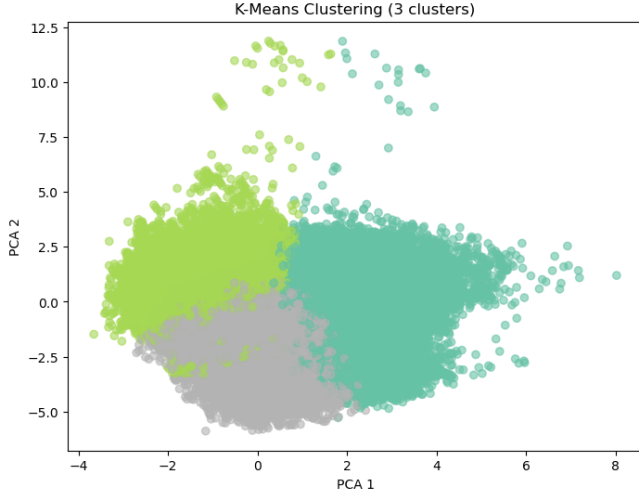


Figure 3: K-means clustering of insurance policies in the PCA-reduced feature space. The clusters show substantial overlap and do not align clearly with claim counts.

sonable and practical definition of claims risk in this setting.

Model performance is evaluated using different metrics depending on the modeling approach. For the decision tree and neural network models, which are trained using mean squared error, performance is assessed using Mean Squared Error (MSE) and the coefficient of determination (R^2). These metrics provide an intuitive measure of prediction accuracy and allow comparison between scratch and library-based implementations.

For the gradient boosting model implemented with LightGBM, a Poisson loss function is used during training, as it is well suited for count data. In this case, Poisson deviance is reported as the primary evaluation metric. In addition, Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) are also computed for the LightGBM model to facilitate direct comparison with the other methods. Finally, the total number of predicted claims is compared to the observed total to assess overall calibration.

It is important to note that the dataset is highly zero-inflated, with the majority of policies having no claims. In such settings, error-based metrics such as Mean Squared Error and R^2 can be misleading, as models that predict values close to zero may achieve low error without capturing meaningful risk patterns. For this reason, MSE and R^2 are primarily used to compare the decision tree and neural network models under a common regression framework. For the LightGBM model, Poisson deviance is emphasized as a more appropriate

metric for count data, as it better reflects performance in the presence of many zero observations.

3.2 Decision Tree Regression

Decision trees are intuitive machine learning models that predict outcomes by recursively splitting the feature space based on simple decision rules. In a regression setting, each split aims to reduce prediction error by grouping observations with similar target values. Decision trees are easy to interpret and can naturally handle non-linear relationships, but they are prone to overfitting if not properly constrained.

In this project, a decision tree regressor is implemented both from scratch and using a standard machine learning library. The scratch implementation follows a CART-style approach, selecting splits that minimize mean squared error and predicting the mean target value in each leaf node. We constrain the model with different parameters such as maximum tree depth and minimum samples per split to control model complexity and reduce overfitting.

The library implementation is used as a baseline to check whether the scratch implementation behaves as expected. By comparing predictions and performance metrics under similar settings, we can verify that the scratch model captures the same overall patterns as the library-based version. While the two models do not produce identical results, differences are expected due to optimized training procedures and implementation details in the library. As we observed similar performance trends, we grew confident in our from-scratch implementation.

Hyperparameters such as maximum tree depth and minimum samples per split were selected through manual experimentation on the training set, aiming to limit overfitting while maintaining interpretability rather than aggressively optimizing performance.

3.3 Neural Network Regression

Feed-forward neural networks are flexible models capable of learning complex, nonlinear relationships through layered transformations of the input features. In a regression context, the network produces continuous outputs by combining weighted inputs through activation functions. The neural network implementation and training procedure follow standard practices described in [2].

A simple neural network regressor with a single hidden layer is implemented from scratch using NumPy.

The network uses the ReLU activation function in the hidden layer and is trained using gradient descent to minimize mean squared error. This implementation provides insight into the training dynamics of neural networks, including forward propagation, backpropagation, and parameter updates.

In addition to the scratch implementation, a reference neural network regressor from a machine learning library is used for comparison. The library based model benefits from optimized training procedures and additional regularization mechanisms. Comparing the two implementations helps assess the correctness of the scratch model and highlights the practical advantages of established libraries when working with real world data.

The number of hidden units, learning rate, and number of training epochs¹ were chosen based on preliminary experiments and validation behavior, prioritizing training stability and simplicity over extensive hyperparameter tuning.

3.4 Gradient Boosting with LightGBM

As a third modeling approach, gradient boosting is applied using the LightGBM framework. Gradient boosting builds an ensemble of decision trees sequentially, where each new tree is trained to correct the errors made by previous trees. This approach is particularly effective for tabular data and is widely used in industrial machine learning applications.

LightGBM is configured to optimize a Poisson loss function, which is well suited for modeling count data such as insurance claim frequency. By directly aligning the training objective with the nature of the target variable, the model can better capture the underlying distribution of claims. Hyperparameters are selected based on cross-validation performance within the training set to balance predictive accuracy and generalization.

Due to the complexity of gradient boosting frameworks, LightGBM was used solely as a library-based implementation.

4 Results

In this section we will analyze and compare the results of our different models.

¹An epoch is one complete pass of the training process over the entire training dataset, during which the model updates its weights to improve predictions.

4.1 Decision Tree Results

The decision tree regressor was evaluated using Mean Squared Error (MSE) and the coefficient of determination (R^2). Both the from-scratch implementation and the library-based implementation achieved similar performance, indicating that the from-scratch implementation behaves as intended.

- **Library implementation**

- MSE: 0.0563
- R^2 : 0.0253

- **From-scratch implementation**

- MSE: 0.0562
- R^2 : 0.0262

The very low R^2 values indicate that the decision tree explains only a small fraction of the variability in claim counts. In practice, this means that the model performs only marginally better than predicting a constant value for all policies. We expected this behaviour as insurance data is highly zero-inflated and most observations have no claims. Simple models struggle to identify meaningful risk patterns in these type of datasets.

Overall, the results suggest that while decision trees can capture limited non-linear structure, a single tree is insufficient to model the complexity of claim frequency data.

4.2 Neural Network Results

The neural network regressors were evaluated using Mean Squared Error (MSE) and the coefficient of determination (R^2), consistent with the decision tree models. Both the library-based and from-scratch implementations produced comparable error magnitudes, although the scratch implementation performed slightly worse.

- **Library implementation**

- MSE: 0.0587
- R^2 : 0.0254

- **From-scratch implementation**

- MSE: 0.0602
- R^2 : -4.41e-05

The low and near-zero R^2 values indicate that the neural networks explain very little of the variance in claim counts. In particular, the negative R^2 obtained by the scratch implementation implies performance slightly worse than a constant baseline model that predicts the mean claim count for all policies.

Despite their increased flexibility compared to decision trees, the neural networks did not yield improved predictive performance. This suggests that model complexity alone is insufficient to address the challenges of insurance claim frequency data, which is dominated by zero observations and affected by substantial noise.

4.3 LightGBM Results

The LightGBM model was evaluated using *Poisson deviance* as the primary metric, along with *RMSE* and *MAE* for comparison with the other models. The model achieved consistently lower Poisson deviance across cross-validation folds, indicating improved performance on count-based predictions.

- **Library implementation**

- RMSE: 0.2336
- MAE: 0.0963
- Poisson Deviance: 0.2889
- Predicted Claims: 28726.4 (28814 total)

In addition, the total number of predicted claims closely matched the observed total, suggesting good overall calibration. Compared to the decision tree and neural network models, LightGBM demonstrated superior predictive performance, reflecting its ability to model non-linear relationships and interactions more effectively in tabular insurance data.

Note that the scale of the error metrics differs from the previous models due to differences in training objectives and prediction behavior under the Poisson loss.

4.4 Comparative Summary

Overall, predictive performance across all models was limited, reflecting the inherent difficulty of insurance claim frequency modeling on highly imbalanced data. Both the decision tree and neural network models achieved low R^2 values and only modest error reductions, indicating that they struggled to capture meaningful variation in claim counts beyond simple baseline behavior.

In contrast, the LightGBM model consistently outperformed the other approaches. When evaluated using Poisson deviance, it demonstrated improved predictive accuracy and better alignment with the observed distribution of claims. In addition, the total number of predicted claims closely matched the observed total, suggesting improved calibration at the portfolio level.

This improvement highlights the importance of aligning the learning objective with the statistical properties of the target variable, particularly in zero-inflated count data.

5 Limitations

While this project provides a comparative analysis of different machine learning approaches for insurance claim frequency modeling, several limitations should be acknowledged.

5.1 Data Characteristics

The dataset is highly imbalanced, with the majority of policies associated with zero claims. This zero-inflation makes it difficult for models to learn informative patterns related to claim occurrence and limits achievable predictive performance. Although this reflects real-world insurance data, it inherently constrains the effectiveness of standard regression models.

In addition, the exposure variable represents a limited observation window of at most one year. While exposure was handled explicitly through capping and feature engineering, modeling longer-term risk dynamics was not possible within the scope of the available data.

5.2 Target Definition and Evaluation

Claims risk was defined as the expected number of claims (*ClaimNb*) during the exposure period rather than a normalized claim frequency. Although this choice is reasonable given the capped exposure and simplifies interpretation, it may obscure finer-grained differences in risk for policies with shorter exposure durations.

Furthermore, evaluation metrics such as MSE and R^2 are known to be less informative in highly zero-inflated settings. While Poisson deviance was used for the LightGBM model to better reflect the count-based nature of the target, the comparability of metrics across models is limited.

5.3 Model Scope

The models explored in our project focus exclusively on claim frequency and do not account for claim severity. In practice, insurance risk assessment typically considers both the likelihood and cost of claims. Additionally, only a limited set of models and feature transformations were examined, reflecting the scope of this study.

5.4 Potential Improvements

The results of this project suggest several concrete directions for improvement. One important limitation is the strong zero-inflation of the target variable. While this was partially addressed by using a Poisson-based objective in the LightGBM model, alternative approaches explicitly designed for excess zeros—such as zero-inflated or hurdle models—could better separate the probability of having any claim from the expected number of claims given that a claim occurs.

Another potential improvement concerns the treatment of exposure. In this project, exposure was capped at one year and incorporated as an input feature. While this choice is reasonable given the structure of the data, modeling claim frequency directly (e.g., *ClaimNb* divided by exposure) or using exposure as an offset in Poisson-based models may provide a more principled way of accounting for time at risk.

Model performance may also benefit from additional feature engineering. In particular, interactions between driver age, vehicle characteristics, and bonus-malus score were not explicitly modeled and may contain useful predictive information. More extensive hyperparameter tuning, especially for the neural network and decision tree models, could also improve performance, although gains are likely limited by the data imbalance.

6 Conclusion and Discussion

In this project, we investigated the use of machine learning methods for predicting the expected number of insurance claims per policy, which corresponds to modeling claim frequency over a one-year exposure period. We implemented and compared a decision tree regressor and a feed-forward neural network, both from scratch and using established libraries, and evaluated them alongside a gradient boosting model based on LightGBM. Claims risk was defined as the expected number of claims during the exposure period, and model performance was assessed using metrics appropriate to the structure of the

data.

The results show that modeling insurance claim frequency is a challenging task due to strong class imbalance and zero-inflation. Both the decision tree and neural network models achieved limited predictive performance, with low R^2 values indicating that they explain only a small fraction of the variability in claim counts. In contrast, the LightGBM model demonstrated improved performance when evaluated using Poisson deviance and produced well-calibrated aggregate predictions, highlighting the importance of aligning the modeling approach and loss function with the statistical properties of count data.

Overall, this study illustrates that increased model complexity alone does not guarantee better performance in insurance risk modeling. Instead, careful consideration of the target definition, evaluation metrics, and data characteristics plays a crucial role. While the models explored here remain far from perfect, the results are consistent with expectations for real-world insurance data and provide insight into the strengths and limitations of different machine learning approaches for claim frequency prediction.

7 References

References

- [1] G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor. *An Introduction to Statistical Learning: With Applications in Python*. Springer, 2023.
- [2] A. Géron. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*. O'Reilly Media, 2nd edition, 2019.
- [3] G. Ke, Q. Meng, T. Finley, et al. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Advances in Neural Information Processing Systems*, 2017.
- [4] F. Pedregosa et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, pp. 2825–2830, 2011.
- [5] A. McCullagh and J. A. Nelder. *Generalized Linear Models*. Chapman and Hall, 2nd edition, 1989.
- [6] OpenAI. *ChatGPT (GPT-based language model) used for programming and debugging assistance*.

[7] Grammarly Inc. Grammarly was used as a writing support tool to assist with grammar, spelling, and clarity of the written report. All content, analysis, modeling decisions, and interpretations were produced by the authors.